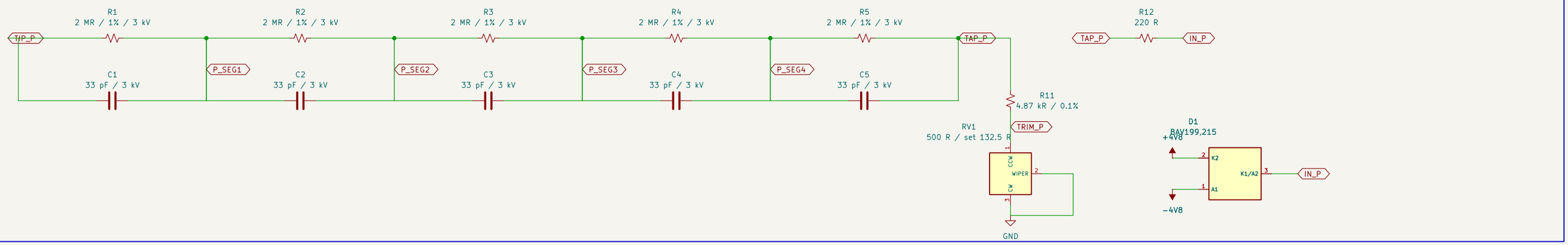
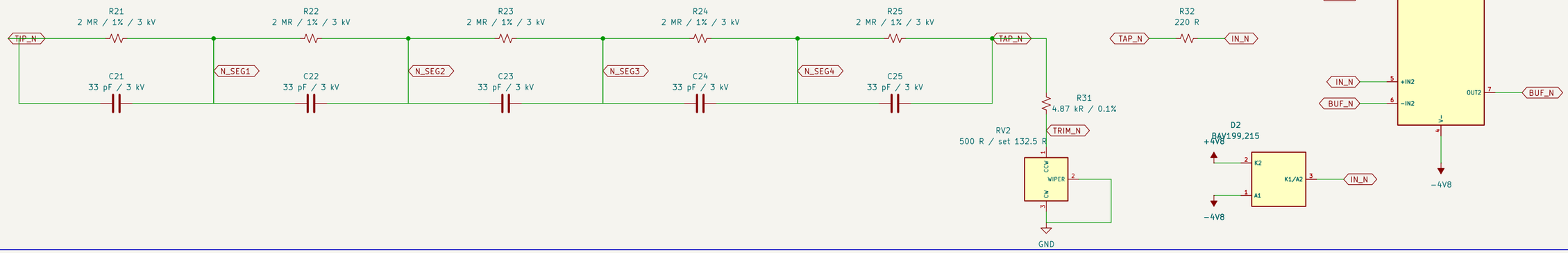
2000:1 INPUTS / mirror-matched geometry about the inter-leg slot, fixed grounded shield

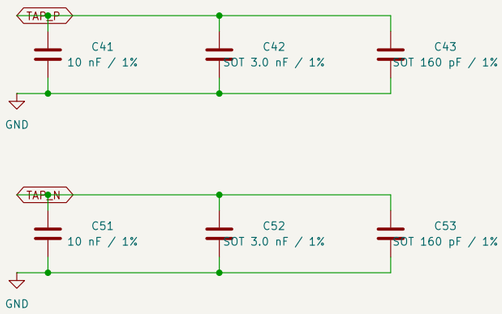
INPUT P | 10 MR + 5.00250 kR | 5 x (2 MR || 33 pF)

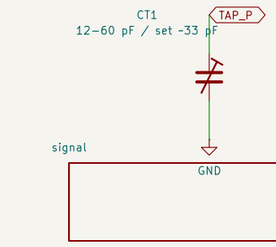


INPUT N | 10 MR + 5.00250 kR | 5 x (2 MR || 33 pF)



AC COMPENSATION / fixed C0G + two select-on-test C0G + accessible fine trim





Fixed 10 nF 1% + select-on-test S1 (E24 1.6-3.9 nF, 1%)
+ S2 (E24 0-360 pF, 1%) + 12-60 pF trimmer.
Nominal 10 nF + 3.0 nF + 160 pF + ~33 pF.
CT1/2: rotor to GND. Adjust through shield tool holes.
TIP_P / TIP_N: captive HV leads.

signal

File: signal.kicad_sch

power

File: power.kicad_sch

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /
File: differential-probe.kicad_sch

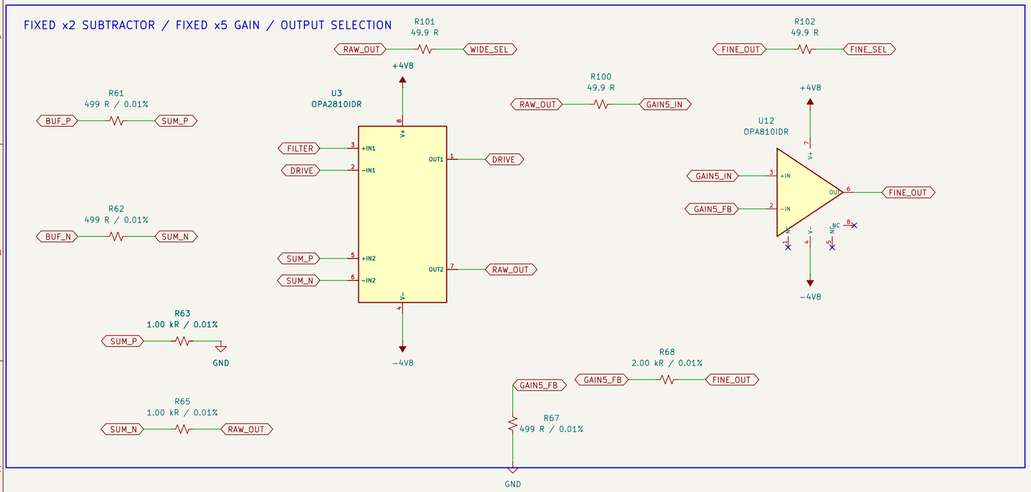
Title: Differential probe / compensated inputs

Size: A3	Date: 2026-10-20	**Rev: C prototype**
KiCad E.D.A. 10.0.4		Id: 1/3

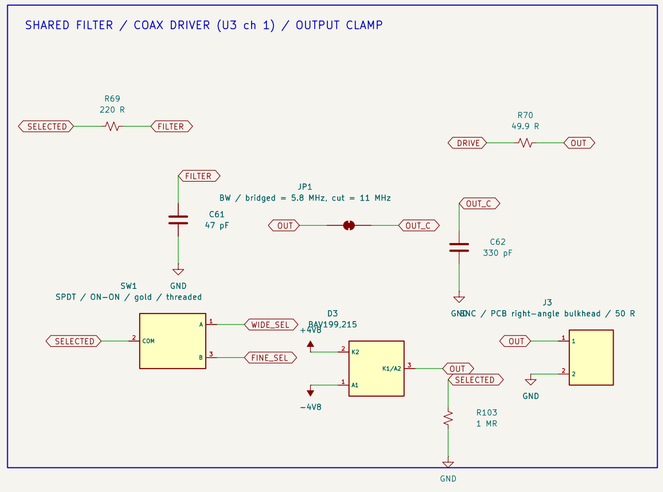
1000:1 / 200:1 OUTPUT / scope input = 1 MR

SW1 common (pin 2) selects 1 = 1000:1, 3 = 200:1. No switched feedback. Verify BBM contacts; allow settling during changes.

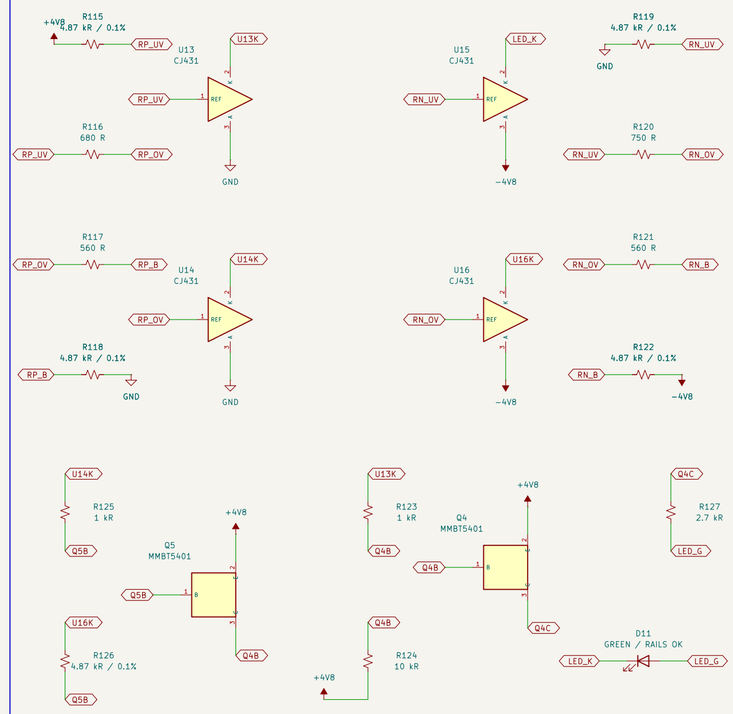
FIXED x2 SUBTRACTOR / FIXED x5 GAIN / OUTPUT SELECTION



SHARED FILTER / COAX DRIVER (U3 ch 1) / OUTPUT CLAMP

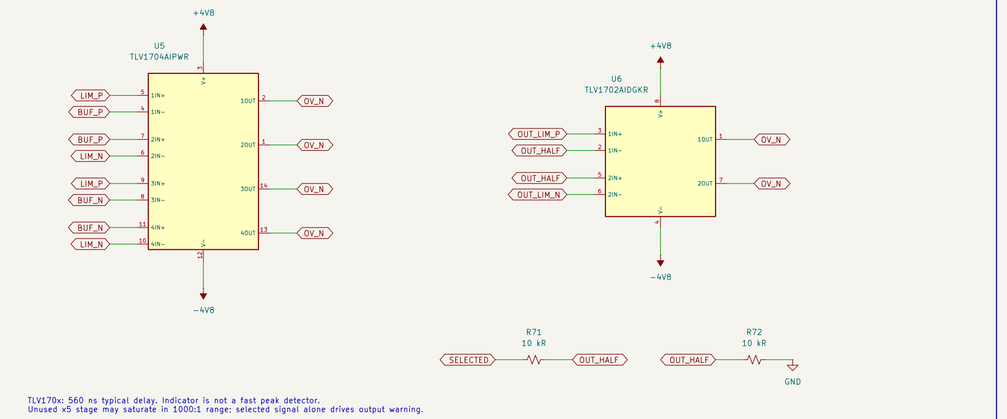


RAILS-OK INDICATOR / green D11 = both rails inside their window



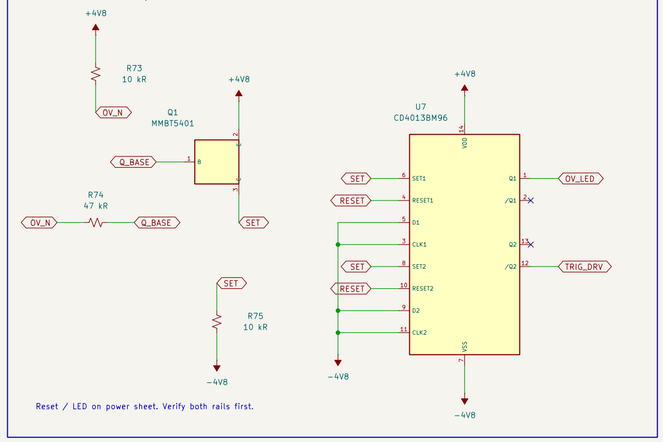
Lit: +rail and −rail each inside its window (validation/checks.txt). Thresholds are not a measurement.

OPERATING-LIMIT WARNING / tips ~1406 V; selected signal ~1.266 V



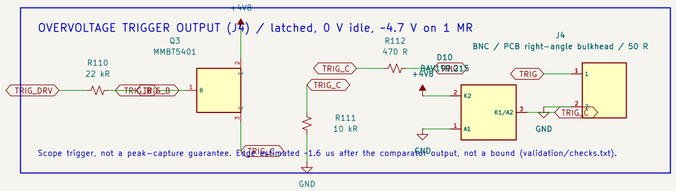
TLV170x: 560 ns typical delay. Indicator is not a fast peak detector.
Unused x5 stage may saturate in 1000:1 range; selected signal alone drives output warning.

LATCHED OVERLOAD / RESET



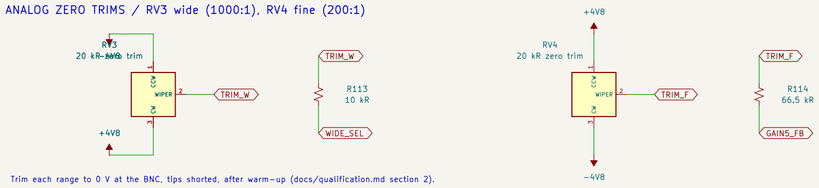
Reset / LED on power sheet. Verify both rails first.

OVERVOLTAGE TRIGGER OUTPUT (J4) / latched, 0 V idle, -4.7 V on 1 MR



Scope trigger, not a peak-capture guarantee. Edge latched ~1.6 us after the comparator output, not a bound (validation/checks.txt).

ANALOG ZERO TRIMS / RV3 wide (1000:1), RV4 fine (200:1)



Trim each range to 0 V at the BNC, tips shorted, after warm-up (docs/qualification.md section 2).

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /signal/
File: signal.kicad_sch

Title: Differential probe / gain, output and overload

Size: A2	Date: 2026-10-20	**Rev: C prototype**
KiCad E.D.A. 10.0.4		Id: 2/3

BATTERY POWER / scope shield defines circuit ground

TWO 9 V BATTERIES / +/−4.8 V LOW-NOISE RAILS

BT1
9 V PCB holder
BAT_P
GND

BT2
9 V PCB holder
GND
BAT_N

SW2
POWER / DPDT / ON-ON / threaded
BAT_P
BAT_N
GND
COM1
COM2
A1
B1
A2
B2
SW_P
SW_N

+BAT
D4
SS14
SW_P

D5
SS14
SW_N
−BAT

C71
10 uF / 25 V
+BAT
GND

C73
10 uF / 25 V
−BAT
GND

U8
ADP7142AUJZ-R7
+BAT
+BAT
VIN
EN
GND
VOUT
ADJ
+4V8
ADJ_0

R81
30 kR
+4V8
ADJ_0

U9
ADP7182AUJZ-R7
−BAT
−BAT
VIN
EN
GND
VOUT
ADJ
−4V8
ADJ_1

R83
29.4 kR
−4V8
ADJ_1

C72
10 uF / 25 V
+4V8
GND

R82
10 kR
ADJ_0
GND

C74
10 uF / 25 V
−4V8
GND

R84
10 kR
ADJ_1
GND

RAIL-REFERENCED COMPARATOR THRESHOLDS (no shunt references)

+4V8
R87
66.5 kR
LIM_P
LIM_P
R88
1 kR
OUT_LIM_P
OUT_LIM_P
R89
10 kR
GND

−4V8
R91
66.5 kR
LIM_N
LIM_N
R92
1 kR
OUT_LIM_N
OUT_LIM_N
R93
10 kR
GND

Thresholds track the LDO rails; budget in tools/budget.rs.
Input clamps D1–D3 return to the rails; rail load absorbs clamp current.

LOCAL DECOUPLING / RAIL REVERSAL DIODES

+4V8
U1
U3
U12
U5
U6
U7
C81 100 nF / 50 V
C83 100 nF / 50 V
C85 100 nF / 50 V
C87 100 nF / 50 V
C89 100 nF / 50 V
C91 100 nF / 50 V
GND
C82 100 nF / 50 V
C84 100 nF / 50 V
C86 100 nF / 50 V
C88 100 nF / 50 V
C90 100 nF / 50 V
C92 100 nF / 50 V
−4V8

+4V8
D6
SS14
GND

D7
SS14
GND
−4V8

RESET / STATUS / POWER VALIDATION

SW3
RESET / NO
+4V8
RESET

R95
100 kR
RESET
−4V8

+4V8
C95
100 nF / 50 V
RESET

−4V8
D8
1N4148W
RESET

R96
10 kR
OV_LED
LED_A

D9
RED / OVERLOAD
LED_A
−4V8

Two 9 V alkaline batteries; center tap = GND.
Verify rails, reset, then connect DUT.
No charging / USB connection during measurement.

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /power/
File: power.kicad_sch

Title: Differential probe / battery supply and thresholds

Size: A3	Date: 2026-10-20	**Rev: C prototype**
KiCad E.D.A. 10.0.4		Id: 3/3